Modelling Orbits and the Motion of a Rocket

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INTRODUCTION

Space exploration has been an ever evolving area or science, ever since us humans first started sending artificial satellites to explore the cosmos in the late 1950s. Any object in space is governed by the laws of motion and universal gravitation discovered by Newton in the late 17th century. Since Sputnik 1, humanity have used these laws to send satellites to space and putting them into intended trajectories that orbit various planets and moons. This report aims to describe a mathematical model designed to simulate the behaviour of an artificial satellite launched to space, orbiting both just the Earth in part (a), and launched to the Moon to take pictures of its surface, whilst keeping a repeatable orbit. Runge-Kutta method of solving ODEs was used when simulating these orbits. [1, 2]

THEORY

Any mass in space is affected by the gravitational force caused by the masses around it, as stated by Newton's law of universal gravitation expressed by *eq.1*.

$$F = m\frac{d^2x}{dt^2} = G\frac{m_1 m_2}{r^2} \tag{1}$$

Where *G* is the gravitational constant, m_1 and m_2 are the masses of the objects and r is the distance between the centre of masses.

In this model's case, a spacecraft, the Explorer orbits the Earth in part (a), where it is only affected by the Earth's gravitational pull and is launched to the Moon in part (b), affected by both the Earth and the Moon's gravitational pull.

Assuming the positions of the Earth and the Moon to be fixed, the equation for the gravitational force caused by these two celestial bodies is expressed in *eq.2* in its vector form.

$$F = m\frac{d^2r}{dt^2} = -\frac{mM_E G}{|r-R_E|^3}(r-R_E) - \frac{mM_M G}{|r-R_M|^3}(r-R_M) \tag{2}$$

Where m is the mass of the spacecraft, M_E and M_M are the masses of the Earth and the Moon respectively, r is the location of the spacecraft.
It is clear to see that *eq.2* is derived from *eq.1*. $(r - R_E)$ and $(r - R_M)$ are the displacement/position vectors that represent the directions of the forces. The expressions in the denominator, $|r - R_E|$ and $|r - R_M|$ represent the magnitudes of the two position vectors.

$$\hat{r_E} = \frac{(r-R_E)}{|r-R_E|}, \hat{r_M} = \frac{(r-R_M)}{|r-R_M|} \tag{3}$$

Eq.3 shows the unit vectors for the two forces, hence the cubed expression in the denominator.

METHOD

For this model, to make calculations simpler, both the Earth and the Moon are assumed to have fixed locations, with the Earth being at the origin and the Moon being at x = 3.844×10^8, y = 0. The function called `explorer_trajectory` was made, in order to solve the ODE using the order Runge-Kutta method using `solve_ivp()` in the SciPy library with the parameter `method='RK45'`.
In `explorer_trajectory`, *eq.2* was broken into its x-y components for both Earth and the Moon parts of the equation. The variables were set by user input but there were also preset variables in case the user did not want to specify the conditions. There were other functions to check the validity of the inputs, e.g. not allowing initial co-ordinates to be within the radius of either body. A similar logic was used to determine whether or not the Explorer experience a crash. Crash functions for each body was created and using the `event=[]` parameter, the integration was stopped whenever the crash conditions were met.
In order to make analysis simpler, the rocket's mass was kept constant, assuming it used all its fuel to get to the launch position (medium Earth orbit, MEO between 6500-7000km), making the rest of the motion cruising around the two bodies, where its position and velocity affected by the gravitational acceleration applied by the two celestial bodies. Atmospheric drag which would affect the velocity of the Explorer was assumed to be zero.
Dry mass of the satellite called the Explorer in this model was given the mass of the Jupiter Icy Moons Explorer, JUICE launched by the European Space Agency. [3]
Gravitational potential energy of the Explorer was calculated using *eq.4*

$$U = -G\frac{Mm}{r} \tag{4}$$

Where M is the mass of the body with the greater mass (Earth or the Moon) and m is the mass of the smaller body (the Explorer), and r is the distance between their centre of mass. The negative sign indicates that the gravity is an attractive force, and does positive work when it attracts a mass. This can be seen in the 'Energy Against Time' plots discussed in the results (*figs.2,4,6,8,10,12,14*). Although using smaller time steps would produce more accurate results, with majority of the graphs produced by this model, the time step (dt) of 10 seconds was used. This is because, the distances and the velocities are in such great magnitudes, taking measurements every 10 seconds did not produce inaccurate results compared to lower time steps like 1-5 seconds, which took slightly longer to run. The relative and absolute error tolerance parameters of the numerical integration method 'RK45' were set to `rtol=1e-7`, and `atol=1e-10` respectively, in order to give accurate results. This model assumes that this is a one body problem for part (a) and two-body problem for part (b), affects of other celestial bodies like the Sun and other planets were not included in the equations. Earth, the Moon and the Explorer are all assumed to be point masses with uniform mass distribution at their centres of mass. External forces like solar winds and magnetic forces are also neglected in this model.

RESULTS AND ANALYSIS

The sizes of the Earth and the Moon are not in scale with the rest of the plots, they are just placed to show where the two bodies are located.

PART A:

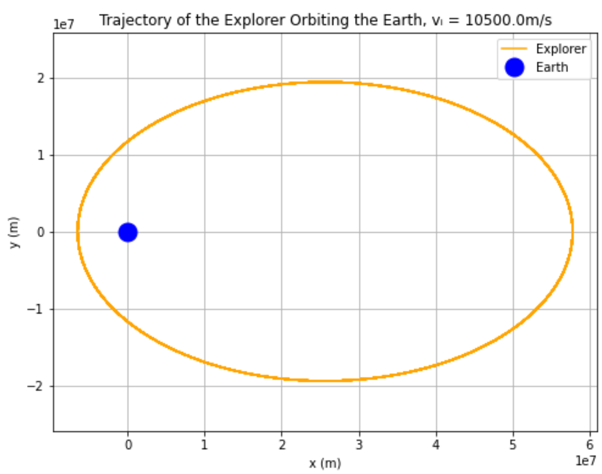

FIG. 1. Position in y against x plot of the Explorer orbiting the Earth with an initial velocity of 10500m/s, launched at x=-6500km, at an angle of 90°. The code was run for 2×10^6s, with time step, dt = 10s. The orbital period is 7.31 days.

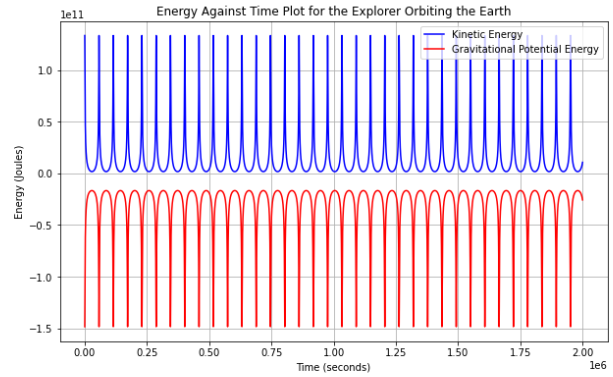

FIG. 2. Change in kinetic and gravitational potential energies against time for the orbit in *fig.1*.

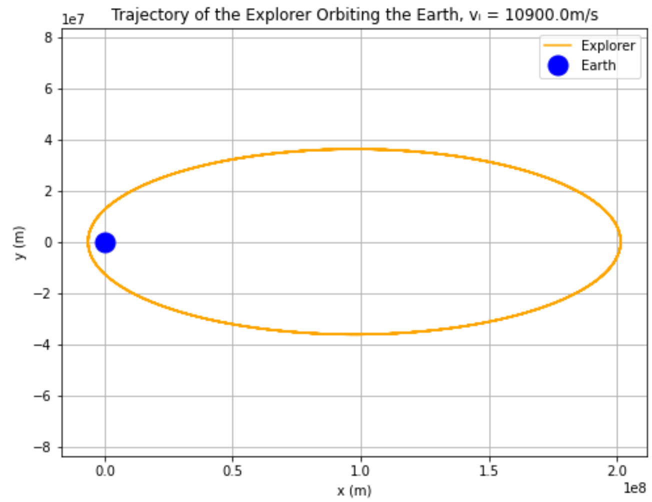

FIG. 3. Position in y against x plot of the Explorer orbiting the Earth with an initial velocity of 10900m/s, launched at x=-6500km, at an angle of 90°. The code was run for 2×10^6s, with time step, dt = 10s. The orbital period is 7.73 days.

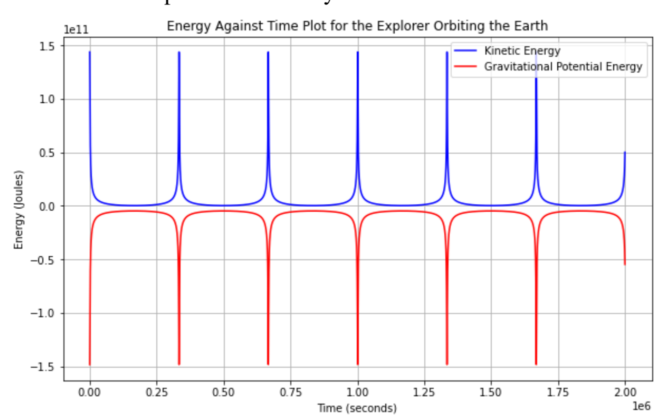

FIG. 4. Change in kinetic and gravitational potential energies against time for the orbit in *fig.3*.

When *figs.1,3* are observed, it is seen that with the same initial launch co-ordinate, the Explorer got further away from Earth when launched at a higher initial velocity. Which meant the orbital period was greater.
The energy changes over time plotted in *figs.2,4*, showed that the kinetic and potential energies were conserved. The peaks show when the Explorer is near the Earth completing a single orbit. A lot more peaks in energy are seen in *fig.2* since the orbit is smaller, the Explorer completes more laps than in *fig.4* over the same period.

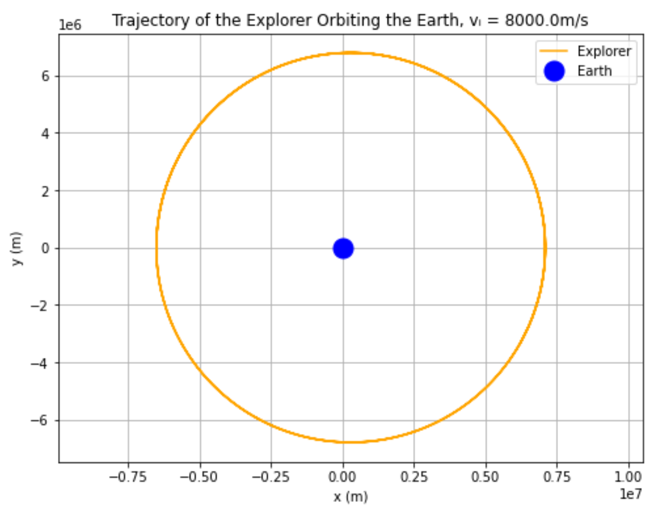

FIG. 5. Position in y against x plot of the Explorer Launched to the Moon with an initial velocity of 8000m/s, launched at x=-6500km, at an angle of 90°. The code was run for $20,000$s, with time step, dt = 10s. The orbital period is 5580s or 93 minutes.

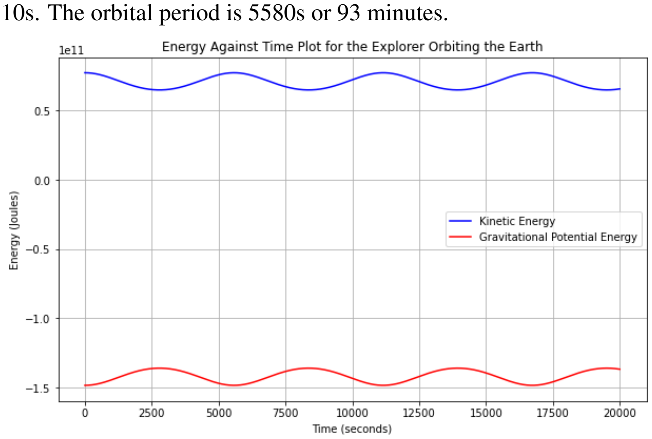

FIG. 6. Change in kinetic and gravitational potential energies against time for the orbit in *fig.5*. The two types of energies are out of phase meaning the conservation of energy still holds.

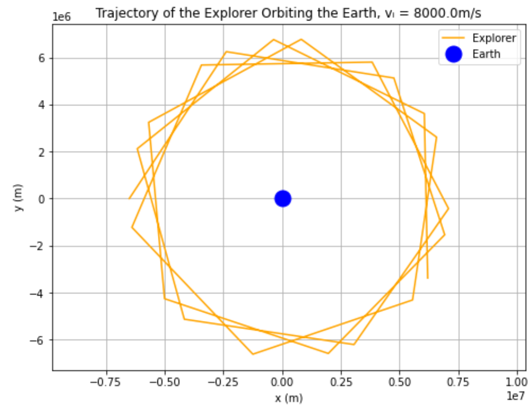

FIG. 7. Position in y against x plot of the Explorer Launched to the Moon with an initial velocity of 8000m/s, launched at x=-6500km, at an angle of 90°. The code was run for $20,000$s, with time step, dt = 1000s. The orbital period is 11000s or about 3 hours.

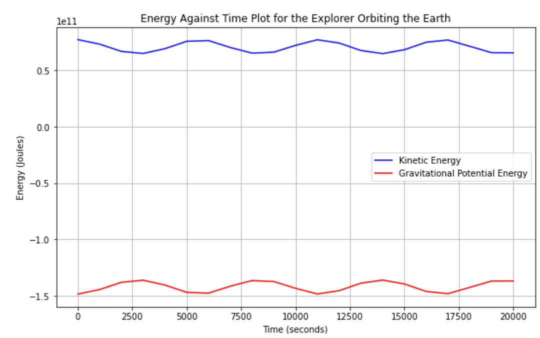

FIG. 8. Change in kinetic and gravitational potential energies against time for the orbit in *fig.7*. The two types of energies are out of phase meaning the conservation of energy still holds.

Whilst a comet-like orbits were observed in *figs.1,3*, in *fig.5* although not perfect, the orbit had a more circular shape. *Figs.7,8* were computed with a time step, dt of 1000s, which caused some inaccuracies since the plots were not as smooth with the reduction in data points. The change in period was quite dramatic compared to *fig.5*. Plots with dt=100s are also included in the appendix, but were not included in this section since they produced the same results as *figs.5,6*.

PART B:

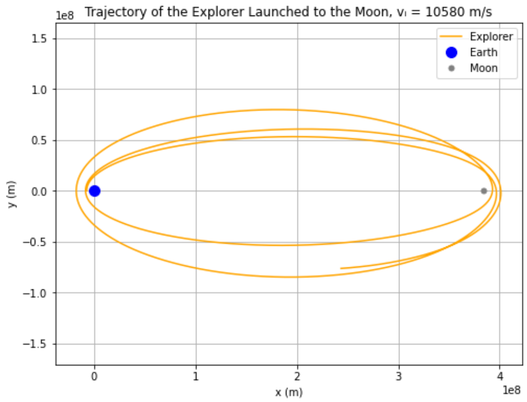

FIG. 9. Position in y against x plot of the Explorer Launched to the Moon with an initial velocity of 10580m/s, launched at x=-7000km, at an angle of 90°. The code was run for 2×10^{6}s, with time step, dt = 10s. The first orbital period is 8.33 days, and closest the Explorer got to the Moon was 8483.49km.

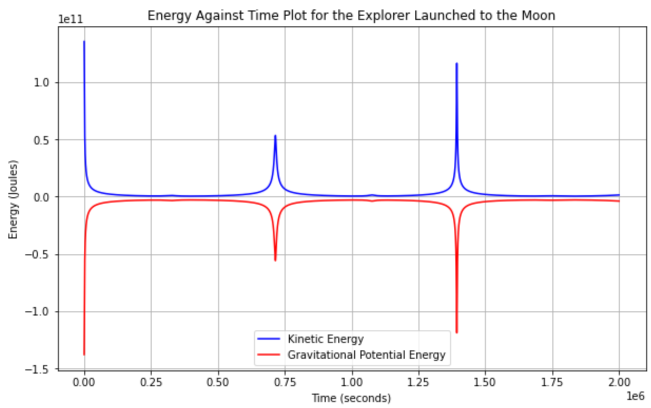

FIG. 10. Change in kinetic and gravitational potential energies against time for the orbit in Figure 9. The greater spike is caused by the Earth and the small one is by the Moon.

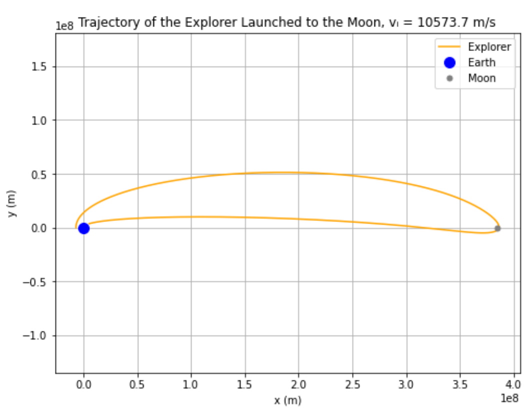

FIG. 11. Position in y against x plot of the Explorer Launched to the Moon with an initial velocity of 10573.7m/s, launched at x=-7000km, at an angle of 90°. The code was run for 2×10^{6}s, with time step, dt = 10s. Closest the Explorer got to the Moon without crashing was 1742.51km, but then it crashed into the Earth.

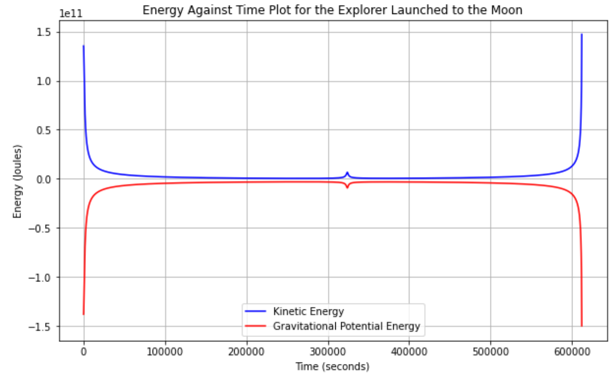

FIG. 12. Change in kinetic and gravitational potential energies against time for the orbit in Figure **??**. The greater spike is caused by the Earth and the small one is by the Moon.

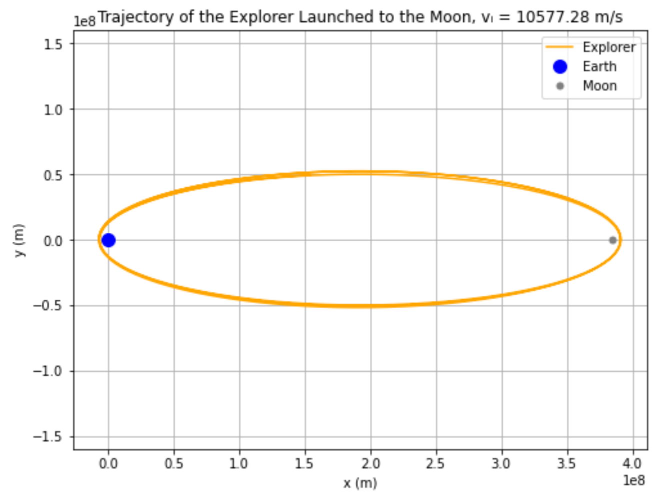

FIG. 13. Position in y against x plot of the Explorer Launched to the Moon with an initial velocity of 10577.28m/s, launched at x=-7000km, at an angle of 90°. The code was run for 2×10^{6}s, with time step, dt = 10s. Without crashing into neither Earth nor the Moon, with a stable orbit, the closest the Explorer got to the Moon was 6523.55km

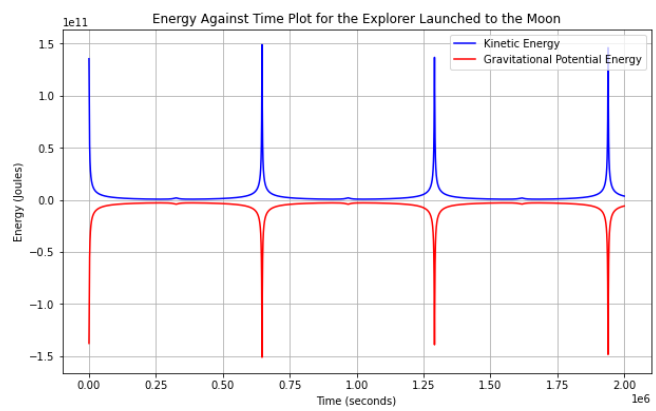

FIG. 14. Change in kinetic and gravitational potential energies against time for the orbit in Figure 17. The greater spike is caused by the Earth and the small one is by the Moon.

DISCUSSION

The computational model simulated the motion of the Explorer relatively accurately within the frame of assumptions made previously. The overall results were independent of the time step but a smaller time step did improve the accuracy of both the plots and the outputs. No visual differences were spotted between dt of 10and 100 seconds but the plots got more rough as dt got bigger than 1000s. The total energy of the Explorer was conserved, shown in 'Energy Against Time' plots below every orbital plot.
The model can definitely be improved by employing a more advanced and quicker programming language, including the affects of other celestial bodies, fields, solar winds, drag, etc.

CONCLUSION

Overall this was a successful attempt at simulating the motion of a spacecraft around the Earth and launched to the Moon and back. SciPy, RK45 method worked fine but since there was no other method of computing the differential equations in this report, we cannot really compare these results with anything other than the data from previous space programs.
Figs.11,13 showed how close we could get to the Moon without crashing into it, although in *fig.11* the satellite crashed into the Earth when it did its full rotation around the Moon, but a stable and repeatable orbit was found in *fig.13*. The closest distances to the moon in these two figures were 1742.51 and 6523.55km respectively.
The conservation of energy between the kinetic and potential energies was also confirmed with what looked like horizontally mirrored images of each other.

REFERENCES

[1] www.nasa.gov. (n.d.). Sputnik. [online] Available at: `www.nasa.gov/history/sputnik/index.html`.

[2] Coates, E.S. (2018). Isaac Newton and the Laws of Motion. The Rosen Publishing Group, Inc.

[3] www.esa.int. (n.d.). Juice. [online] Available at: `www.esa.int/Science_Exploration/Space_Science/Juice`.

APPENDIX

Name	Value
r_{Earth}	6371×10^6 m
r_{Moon}	1.7374×10^6 m
M_{Earth}	5.972×10^{24} kg
M_{Moon}	7.342×10^{22} kg
$m_{Explorer}$	2420 kg
G	6.6743×10^{-11} m$^3 kg^{-1} s^{-1}$
x_{Moon}	3.844×10^8 m

TABLE I. Table of variables/constants used in the model. x_{Moon} is the distance between the Earth and the Moon.

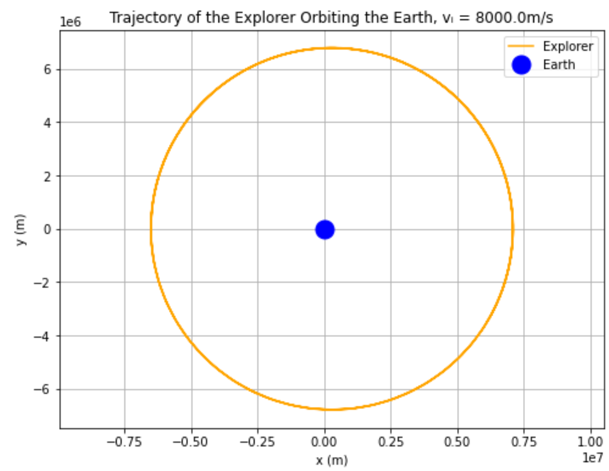

FIG. 15. Position in y against x plot of the Explorer Launched to the Moon with an initial velocity of 8000m/s, launched at x=-6500km, at an angle of 90°. The code was run for 20,000s, with time step, dt = 100s. The orbital period is 5580s or 93 minutes.

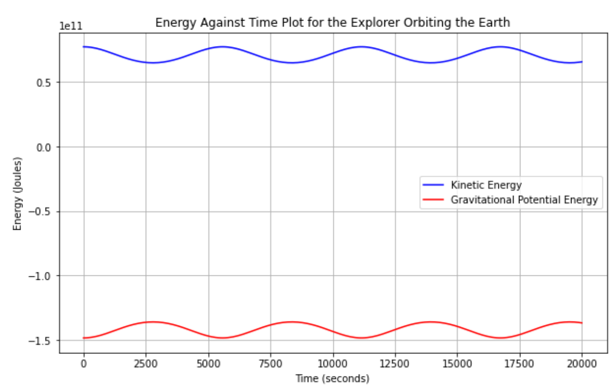

FIG. 16. Change in kinetic and gravitational potential energies against time for the orbit in *fig.15*. The two types of energies are out of phase meaning the conservation of energy still holds.

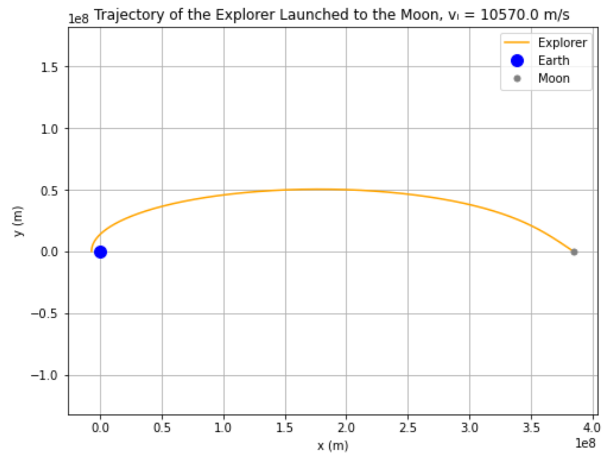

FIG. 17. Position in y against x plot of the Explorer Launched to the Moon with an initial velocity of 10570m/s, launched at x=-7000km, at an angle of 90°. The code was run for 2×10^6s, with time step, dt = 10s. It crashed to the moon at t=329437.35s, in around 3 days and 20 hours.

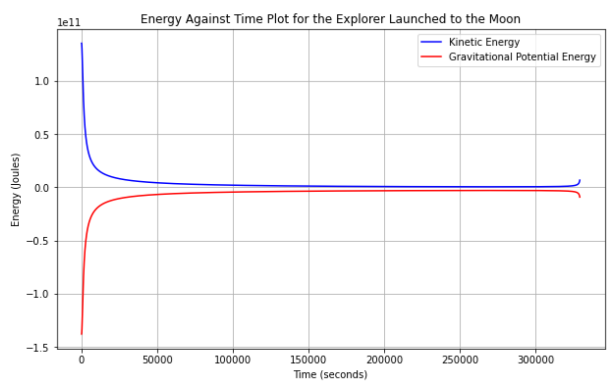

FIG. 18. Change in kinetic and gravitational potential energies against time for the launch in *fig.17*. The two types of energies are out of phase meaning the conservation of energy still holds.